Actual first crossing-jet factors

Base-content bridge and one precision layer, 14 November 2026

Source `e422291022ee713b7097438f322baa4a0bf794f2`

Result scope. The actual completed orientation kernel has explicit weighted first-jet factors with 2^u labels. The residual is one bit above the actual diagonal weight. The actual base chart's integrality is derived through a literal polynomial identity to an existing construction. These are source-bound component claims; all PCC records are unsigned [pcc:unverified].

The same deformation. Let $L = 2^a$, $S = \{-L, \dots, L-1\}$, $C(T) = \prod_{s\in S}(T-s)$, $\kappa = C'(0)$ and $\alpha = 2^{a+1}$. For actual ports $0 \le x, y < L$, set $\delta = y - x$ and

$$\begin{aligned} M_{xy} &= \{j < u : \mathrm{bit}_j(x) = \mathrm{bit}_j(y)\},\\ B_\delta(T) &= \kappa^{-1} \prod_{s\in S\setminus\{\delta\}} (T-s),\\ F_{xy}(T) &= \prod_{j\in M_{xy}} \left(\frac{2^j - T}{2^j - \delta}\right)^{m_j}, \qquad W_{xy}(T) = B_\delta(T) F_{xy}(T),\\ G_i &= m_i - \sum_{i<j<u} (j-i) m_j. \end{aligned}$$

The first-layer theorem requires $i < u \le a$ and $m_j > 0$ for every retained j. Global chart integrality uses feasible $G_j \ge 0$ at every retained level. All previous source values, origin identity and degree bounds apply to this same W; it is not replaced separately at each level.

Actual base-content bridge. Prior normalizer and common-scalar sources already construct the actual local base polynomial over the concrete rational subring

$$\mathbb{Z}_{(2)} = \{q : \mathrm{padicNorm}_2(q) \le 1\}.$$

Their actual finite-product normalizer unit is derived from Lucas parity and factorial identities. These prior results are reused, not claimed as new.

The new bridge proves the exact grid bijection $k \mapsto k - L$, transports punctured products and identifies the derivative normalization with the prior center product. It then proves a literal rational-polynomial identity

$$\mathrm{map}(\mathrm{subtype})(\mathrm{localBase}_{m,s}) = B_{\mathrm{node}(s)}\big(\mathrm{node}(m) + \alpha Z\big).$$

Equality uses every rational evaluation over an infinite field, not a finite sample. Mapping actual coefficients gives the desired base-content bound in every literal signed source chart, including zero and $-L$. Multiplication by the actual correction yields whole-kernel chart integrality. The old exposed base-content premise is no longer supplied. PCC-20261114-bbdb2c/ab220a [pcc:unverified].

Content convention. The predicate $\mathrm{VBound}(k, q)$ means $q = 0$ or $v_2(q) \ge k$. It therefore handles zero without using the valuation API's default zero value as a finite valuation bound.

Actual diagonal and normalized residues

Put $t = -2^i$, and let J_{xy} be coefficient one of the actual polynomial $W_{xy}(t + \alpha Z)$. The diagonal correction value is

$$D_i = F_{00}(t) = \prod_{j<u} \left(\frac{2^j + 2^i}{2^j} \right)^{m_j} > 0.$$

The numerator valuation is $i + 1$ for $j = i$, and $\min(i, j)$ otherwise. Consequently $v_2(D_i) = G_i$ exactly. A lower bound alone would not justify dividing the strict margin by this diagonal.

Actual first coefficient. Differentiate the punctured-polynomial identity at the actual source node. A crossing has equal bit i, hence $\delta \neq t$, and $B_\delta(t) = 0$. The derivative product rule gives

$$\Gamma = \alpha C'(t)/\kappa, \qquad J_{xy} = \frac{\Gamma}{t - \delta} F_{xy}(t),$$
$$c = J_{00} = \frac{\Gamma}{t} D_i, \qquad \frac{J_{xy}}{c} = \frac{t}{t - \delta} \frac{F_{xy}(t)}{D_i}.$$

The actual derivative ratio equals the prior actual cauchyRho, whose valuation is zero. Thus $\Gamma \neq 0$, $v_2(\Gamma) = a + 1$, $c \neq 0$, and the exact public scalar weight is

$$w = v_2(c) = a + 1 - i + G_i.$$

No favorable scalar, unspecified unit or desired rank is supplied. PCC-20261114-e9e4c2/bdaedb [pcc:unverified].

Matching retained classes. If $x \equiv y \pmod{2^u}$, every retained bit agrees and the actual mask is the full retained range. Then

$$\frac{F_{xy}(t)}{D_i} = \prod_{j<u} \left(\frac{2^j}{2^j - \delta} \right)^{m_j} \equiv 1 \pmod 2.$$

Each ratio is the inverse of $1 - \delta/2^j$ with strictly even perturbation. The base ratio $t/(t - \delta)$ also equals one modulo two since $i < u$. Their product gives $J_{xy}/c \equiv 1 \pmod 2$.

Nonmatching retained classes. The actual first differing bit s is below u, omitted by the mask, and $s \neq i$ at a crossing. If $s < i$, anchor i contributes extra $(i - s)m_i$. If $i < s < u$, omitting anchor s avoids loss $(s - i)m_s$. Positivity of all retained multiplicities gives at least one extra bit. The actual correction chart has content $G_i + 1$; its constant coefficient therefore does too. Dividing by D_i leaves content one. The actual base ratio is integral because $v_2(t - \delta) \leq i$. Hence $J_{xy}/c \equiv 0 \pmod 2$ outside a matching retained class. These are quantified source proofs [pcc:unverified].

Explicit weighted factors

Zero out noncrossing entries and call the normalized first-jet matrix R. For labels $r \in \{0, \dots, 2^u - 1\}$, define actual rational 0/1 matrices

$$U_{xr} = [x \bmod 2^u = r], \qquad V_{ry} = [y \bmod 2^u = r].$$

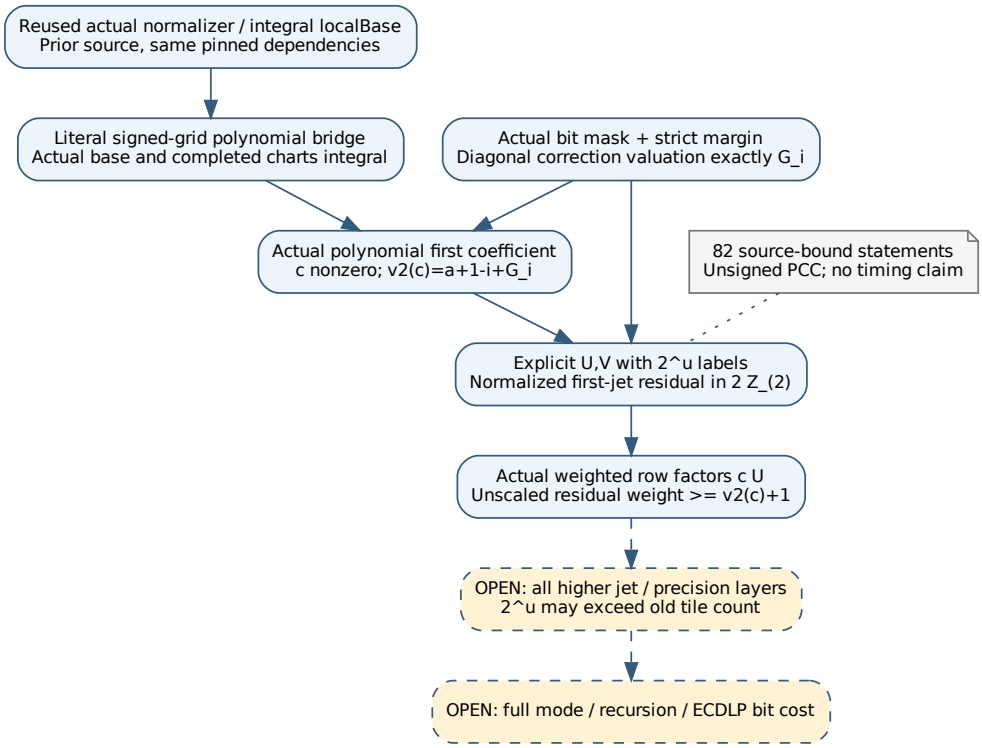
The exact finite matrix product is $(UV)_{xy} = [x \bmod 2^u = y \bmod 2^u]$. Both factors are integral. The actual kernel analysis on page2 gives

$$\mathrm{VBound}\big(1, R_{xy} - (UV)_{xy}\big).$$

For actual weighted row factors $A = cU$ and actual unnormalized crossing first-jet matrix J_{cross},

$$\mathrm{VBound}(w, A_{xr}), \qquad \mathrm{VBound}\big(w+1, (J_{\mathrm{cross}} - AV)_{xy}\big).$$

Feasibility $G_i \geq 0$ makes weighted row factors integral as well. Every entry, normalization and error is derived from the same actual polynomial family. No factorization oracle or assumed low-rank matrix appears. PCC-20261114-800c08/f93c33/9ba2ee [pcc:unverified].



Cost qualification. The count 2^u covers both bit classes in the full port crossing matrix. It may exceed the old 2^{i+1} tile count. It is not the rank or factor count of the entire higher-precision/higher-jet kernel. No asymptotic speedup follows from this one layer; all residual layers, source modes, precision, degree, recovery and recursion still need complete accounting. Additive weights must remain additive.

Evidence and unresolved algorithmic obligations

Formal artifacts. Six new modules contain 82 authored statements. Frozen source: `e422291022ee713b7097438f322baa4a0bf794f2`.

The pinned shared axiom gate audits 82 declarations with 0 extra trusted axioms. Import-inclusive fresh OrientationLeading exits0 with default imports. New theorem records are source-bound and unsigned; producer checks do not manufacture signed standing [pcc:unverified].

The proof map includes every exact declaration, quantifier and hypothesis. Source/dependency hashes retain the prior normalizer and common-scalar sources. Failed attempts and raw outputs are preserved. The earlier 66 orientation statements and their claims remain unchanged; the old exposed base-content dependency is addressed by the new bridge.

Literal identity artifact. Certificate `IDC1h57849461da3c8854` checks only $(N - UV)c = cN - c(UV)$ over the schema's field, at 32+16 points. It does not certify masks, nodal derivatives, valuations, matrix sums, rank or expected cost. Those quantified statements use Lean. The 3 new adapter-library tests, formatting and strict Clippy pass. No cryptographic fixture, solve, performance/scaling run or old native panel was launched [pcc:unverified].

Primary sources. KB unavailable; bounded fallback queries and primary openings are recorded in the protocol/journal. Read actual `LocalCharts`, `LocalIntegral` and the normalized-square normalizer before implementation, with matching Git blob IDs. Read pinned Oct25 D.4.3/D.4.5 for the original kernel/mode interface and the previous repeated-anchor/common-kernel notes for counterexamples and retained-u Fibonacci costs. Reused arithmetic is not a novelty claim; a bounded search does not establish priority.

Requirement status.

Actual base-content binding	Constructed literal polynomial bridge
Actual scalar and first jet	Nonzero diagonal and exact weight derived
First-jet lowest precision layer	Explicit integral factors and residual
Pinned source axiom audit	82 statements,0 extra trusted axioms
Fresh replay	Exit0, default imports
All higher jets/precision layers	Open
Expected ECDLP bit cost/existence	Open
PR integration	Protected concurrent ledger conflict; no reconciliation

Next discriminating work. Extend the actual factorization to the source's required chart precision and jets, with every new direction and weight charged. Then prove uniform packet/child bounds, immutable entry costs, recovery and complete recursion. A full ECDLP statement also needs the charged public-input reduction/finder/extraction and expected bit cost. These targets are not completed by a first-layer component.

`research/prime_orientation_layers_20261114/PROOF_MAP.md`
`research/prime_orientation_layers_20261114/evidence/source-freeze.json`
`research/prime_orientation_layers_20261114/evidence/claim-pins.json`

The diagram is explanatory. Canonical performance/solve graphs contain no new value from this round.